



Exploring Structured Semantic Priors Underlying Diffusion Score for Test-time Adaptation

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Highlights

We propose **DUSA**, a competitive test-time adaptation method developed to enhance discriminative tasks using generative diffusion models, with improved efficiency.

- ✓ A **novel proposition** is presented to uncover the underlying **semantic structure** of score-based conditional diffusion models **at any single timestep**.
- ✓ A **test-time objective** is introduced to utilize the **implicit semantic priors**, enhancing the task model's adaptation **efficiency** while boosting its **performance**.
- ✓ **Several practical designs** are advocated and validated to improve adaptation **efficiency** while preserving training **stability**.

Problem Formulation

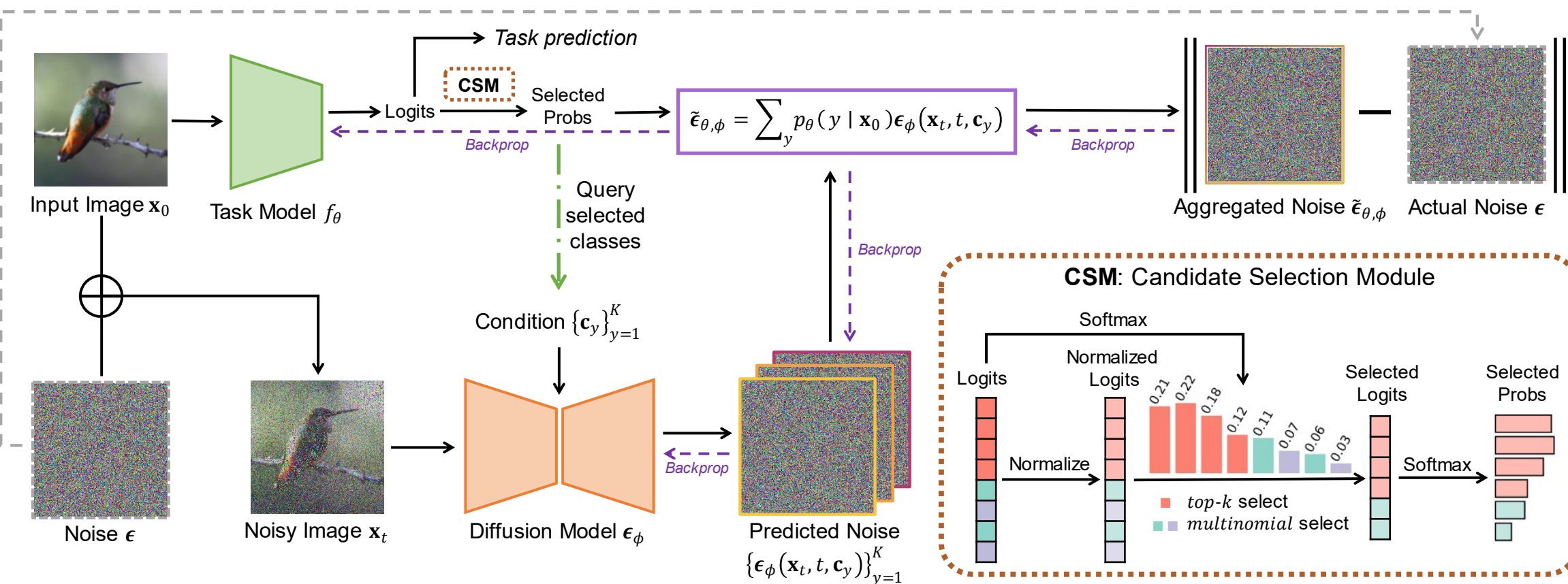
In Test-time Adaptation (TTA), a task model f_θ is pre-trained on labeled source data $\mathcal{D}_S = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$. During test time, the source data is **no longer accessible** and we aim at improving the task model f_θ on the **unlabeled** target data $\mathcal{D}_T = \{\mathbf{x}_j\}_{j=1}^M$ **on the fly**.

Motivations

- ✓ Generative modeling provides a more **comprehensive** and **robust** understanding of data, yielding them strong candidates to facilitate the task of test-time adaptation.
- ✓ Existing art with diffusion modeling is reliant on applying the Monte-Carlo method in the conditioning space, which could be further improved in **efficiency** and **versatility**.

Framework

- ✓ With the **implicit priors** unveiled, we propose a **test-time objective** to enable the task model f_θ to **extract knowledge** from the semantic structure of the diffusion model ϵ_ϕ .
- ✓ Per-class noise predictions ϵ_ϕ are ensembled based on probabilities p_θ from the task model, and the aggregated noise $\tilde{\epsilon}_{\theta,\phi}$ is then aligned with the true noise ϵ .
- ✓ To further reduce computational overhead, a Candidate Selection Module is devised.



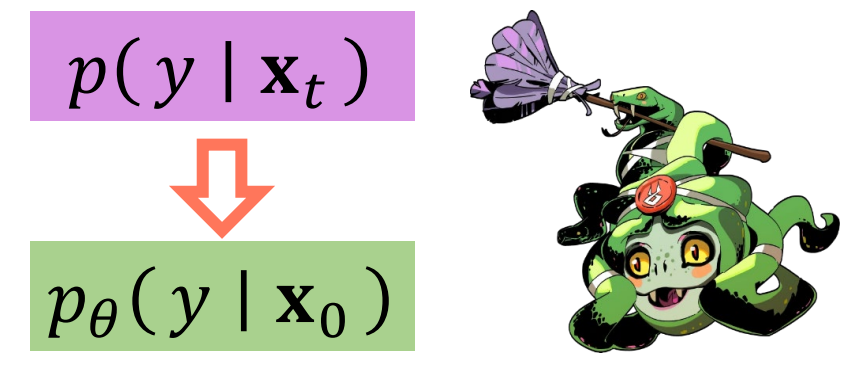
Diffusion Score for Test-time Adaptation

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = \sum_y \underbrace{p(y | \mathbf{x})}_{\text{score function}} \underbrace{\nabla_{\mathbf{x}} \log p(\mathbf{x} | y)}_{\text{implicit priors}} \underbrace{\epsilon}_{\text{cond. score functions}}$$

$$\text{Score-Noise Connection} \quad \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) = -\frac{\epsilon}{\sqrt{1 - \alpha_t}}$$

$$\epsilon = \sum_y \underbrace{p(y | \mathbf{x}_t)}_{\text{real noise}} \underbrace{\epsilon_\phi(\mathbf{x}_t, t, \mathbf{c}_y)}_{\text{implicit priors}} \underbrace{\epsilon_\phi(\mathbf{x}_t, t, \mathbf{c}_y)}_{\text{cond. noise estimations}}$$

Guide task model with implicit semantic priors



$$\mathcal{L}_{DUSA} = \mathbb{E}_\epsilon \left[\left\| \epsilon - \sum_y p_\theta(y | \mathbf{x}_0) \epsilon_\phi(\mathbf{x}_t, t, \mathbf{c}_y) \right\|_2^2 \right]$$

Versatility

Diverse Diffusion Training Objectives

- ✓ Our **DUSA** is applicable to a **wide range** of diffusion models trained on ϵ -prediction, $\mathbf{x}_0$ -prediction, $\mathbf{v}$ -prediction, and other similar objectives.

Free Lunch in Modern Diffusion Models

- ✓ In modern CFG-based diffusion models, an additional **semantic structure** emerges:

$$\epsilon_\phi(\mathbf{x}_t, t, \phi) = \sum_y p(y | \mathbf{x}_t) \epsilon_\phi(\mathbf{x}_t, t, \mathbf{c}_y)$$

- ✓ Thus, the **DUSA** objective can be more **efficient** with minimal performance drop:

$$\mathcal{L}_{cond}(\theta) = \mathbb{E}_\epsilon \left[\left\| \epsilon - \sum_y p_\theta(y | \mathbf{x}_0) \epsilon_\phi(\mathbf{x}_t, t, \mathbf{c}_y) \right\|_2^2 \right], \mathcal{L}_{uncond}(\phi) = \mathbb{E}_\epsilon \left[\left\| \epsilon - \epsilon_\phi(\mathbf{x}_t, t, \phi) \right\|_2^2 \right]$$

$$\mathcal{L}_{DUSA-U}(\theta, \phi) = \mathcal{L}_{cond}(\theta) + \mathcal{L}_{uncond}(\phi)$$

Readily Applicable to Dense Prediction Tasks

- ✓ DUSA handles dense tasks like semantic segmentation while keeping **simplicity**.

$$\mathcal{L}_{DUSA-seg}(\theta, \phi) = \mathbb{E}_{\epsilon, (h, w)} \left[\left\| \epsilon - \sum_{k=1}^K p_\theta(y | \mathbf{x}_0)_{h, w, k} \cdot \epsilon_\phi(\mathbf{x}_t, t, \mathbf{c}_k)_{h, w} \right\|_2^2 \right]$$

Experiments

Fully TTA of ImageNet Pre-trained Classifiers

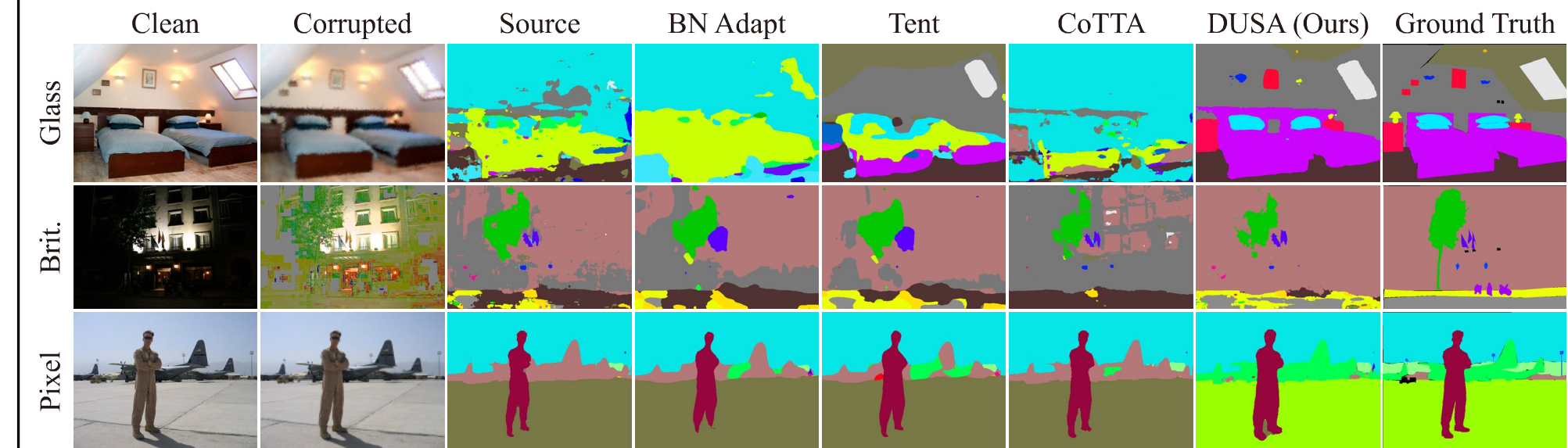
Method	Noise			Blur				Weather				Digital				Avg.
	Gauss.	Shot	Impul.	Defoc.	Glass	Motion	Zoom	Snow	Frost	Fog	Brit.	Contr.	Elastic	Pixel	JPEG	
ResNet-50 (GN)	22.1	23.0	22.0	19.8	11.4	21.5	25.0	40.3	47.0	34.0	68.8	36.3	18.5	29.3	52.6	31.4
• Tent	25.3	29.1	24.5	14.9	9.9	21.6	22.3	27.5	32.1	3.5	69.9	42.0	10.3	48.6	54.6	29.1
• CoTTA	22.1	23.0	22.0	19.8	11.4	21.5	25.1	40.3	47.0	34.0	68.8	36.4	18.5	29.3	52.6	31.5
• EATA	38.6	40.9	39.7	27.3	26.7	36.5	38.6	50.8	49.1	55.6	72.0	49.9	40.5	55.7	58.2	45.3
• SAR	39.6	42.4	41.0	19.8	22.9	37.1	38.7	27.3	47.4	55.1	72.4	48.8	7.2	54.9	57.4	40.8
• RoTTA	22.8	23.8	22.5	19.7	12.0	21.8	25.2	41.3	47.5	34.6	69.2	36.8	19.2	29.9	52.9	31.9
• Diffusion-TTA	42.0	44.6	42.4	38.3	39.5	46.9	48.2	56.5	56.3	60.0	72.6	45.6	57.9	61.4	58.0	51.3
• DUSA (Ours)	45.2 _{±0.0}	47.3 _{±0.0}	46.3 _{±0.1}	37.3 _{±0.1}	37.6 _{±0.2}	48.4 _{±0.3}	50.3 _{±0.3}	59.1 _{±0.1}	55.6 _{±0.0}	63.3 _{±0.3}	73.3 _{±0.1}	55.1 _{±0.0}	56.5 _{±0.3}	63.2 _{±0.1}	60.9 _{±0.2}	53.3
• DUSA-U (Ours)	45.0 _{±0.1}	47.1 _{±0.1}	46.1 _{±0.0}	36.8 _{±0.2}	37.7 _{±0.1}	47.9 _{±0.1}	49.5 _{±0.3}	59.0 _{±0.1}	55.4 _{±0.1}	63.0 _{±0.2}	73.1 _{±0.1}	54.3 _{±0.0}	56.4 _{±0.2}	62.9 _{±0.1}	60.5 _{±0.3}	53.0
ViT-B/16 (LN)	38.3	35.4	38.1	29.5	24.2	32.8	30.5	36.4	45.0	50.4	68.3	22.5	39.4	52.7	53.5	39.8
• Tent	53.9	54.5	54.1	44.4	47.2	53.8	6.7	4.6	61.9	65.4	72.9	54.9	58.0	65.1	64.1	50.8
• CoTTA	38.3	35.4	38.1	29.5	24.2	32.8	30.5	36.4	45.0	50.4	68.3	22.5	39.4	52.7	53.5	39.8
• EATA	55.4	56.3	55.3	48.9	53.4	58.6	58.2	63.5	64.1	67.5	74.3	56.5	65.7	68.5	66.6	60.9
• SAR	53.9	54.3	54.1	46.0	47.8	54.2	49.4	28.2	61.4	64.3	72.8	54.3	59.2	64.8	63.5	55.2
• RoTTA	42.6	39.9	42.9	30.6	26.4	34.8	31.7	39.2	47.8	52.4	68.8	23.3	42.0	55.0	54.0	42.1
• Diffusion-TTA	52.1	54.5	53.5	49.3	52.9	56.9	55.6	60.6	63.0	64.2	72.6	47.4	66.4	67.6	62.5	58.6
• DUSA (Ours)	56.6 _{±0.2}	57.9 _{±0.2}	57.0 _{±0.0}	53.3 _{±0.1}	56.7 _{±0.3}	62.4 _{±0.1}	61.6 _{±0.1}	65.9 _{±0.1}	65.7 _{±0.1}	70.1 _{±0.1}	75.3 _{±0.1}	60.2 _{±0.3}	67.9 _{±0.1}	69.7 _{±0.1}	65.8 _{±0.1}	63.1
• DUSA-U (Ours)	56.3 _{±0.1}	57.6 _{±0.1}	56.7 _{±0.1}	52.5 _{±0.1}	56.4 _{±0.3}	61.9 _{±0.1}	60.4 _{±0.2}	65.8 _{±0.2}	65.4 _{±0.2}	70.0 _{±0.1}	75.3 _{±0.0}	58.7 _{±0.2}	67.8 _{±0.1}	69.4 _{±0.0}	64.3 _{±0.1}	62.6
ConvNeXt-L (LN)	56.7	56.2	58.3	35.1	20.7	47.6	43.5	58.9	59.8	48.0	76.6	55.7	34.0	42.3	63.3	50.5
• Tent	57.4	57.8	58.9	35.7	24.3	51.3	46.3	59.8	58.4	11.0	77.1	61.2	35.1	50.0	64.4	49.9
• CoTTA	56.7	56.2	58.3	35.1	20.7	47.6	43.5	59.0	59.9	48.0	76.6	55.7	34.0	42.3	63.3	50.5
• EATA	57.5	58.0	59.0	38.7	27.1	51.6	47.0	60.7	58.5	49.3	77.2	61.3	40.2	50.3	64.5	53.4
• SAR	57.0	56.7	58.8	37.4	26.6	50.9	46.3	60.1	57.6	12.4	77.0	61.9	37.1	51.4	64.1	50.4
• RoTTA	57.0	56.7	58.7	35.1	21.3	48.0	44.0	59.5	60.0	48.9	76.6	56.8	34.6	43.1	63.4	50.9
• Diffusion-TTA	58.7	59.6	58.3	50.3	48.8	57.6	54.8	63.3	64.8	68.6	77.4	60.9	62.0	65.6	65.5	61.1
• DUSA (Ours)	64.2 _{±0.1}	65.5 _{±0.1}	65.6 _{±0.1}	54.7 _{±0.1}	53.6 _{±0.2}	63.8 _{±0.1}	61.9 _{±0.1}	70.1 _{±0.1}	66.6 _{±0.2}	72.7 _{±0.3}	79.7 _{±0.0}	68.9 _{±0.0}	66.1 _{±0.2}	70.7 _{±0.2}	69.3 _{±0.1}	66.2
• DUSA-U (Ours)	63.8 _{±0.1}	65.2 _{±0.0}	65.2 _{±0.1}	54.0 _{±0.1}	53.3 _{±0.2}	63.3 _{±0.1}	60.6 _{±0.1}	69.9 _{±0.1}	66.4 _{±0.1}	72.5 _{±0.2}	79.6 _{±0.0}	68.1 _{±0.0}	65.9 _{±0.2}	70.3 _{±0.2}	68.7 _{±0.1}	65.8

Continual TTA of ImageNet Pre-trained Classifiers

Time	t →															
Method	Gauss.	Shot	Impul.	Defoc.	Glass	Motion	Zoom	Snow	Frost	Fog	Brit.	Contr.	Elastic	Pixel	JPEG	Avg.
ConvNeXt-L (LN)	56.7	56.2	58.3	35.1	20.7	47.6	43.5	58.9	59.8	48.0	76.6	55.7	34.0	42.3	63.3	50.5
• Tent	57.4	60.0	62.9	38.7	32.8	53.7	50.0	60.3	60.2	67.4	77.5	64.9	23.4	52.3	64.6	55.1
• CoTTA	56.7	56.2	58.3	35.1	20.7	47.6	43.5	59.0	59.9	48.1	76.6	55.8	34.1	42.3	63.3	50.5
• SAR	57.0	59.6	62.6	40.9	32.5	55.1	51.1	61.1	61.2	68.3	78.0	65.4	28.4	52.1	65.2	55.9
• EATA	57.6	61.0	63.5	42.5	35.2	55.3	52.4	62.3	62.9	68.6	78.3	66.1	46.2	56.7	66.9	58.3
• RoTTA	57.0	58.2	60.9	34.2	24.5	47.9	45.3	60.9	62.5	51.7	74.9	49.8	39.3	42.6	62.5	51.5
• Diffusion-TTA	58.1	63.2	63.2	54.1	56.6	61.8	62.5	65.2	65.5	68.1	75.3	58.9	37.3	54.8	60.9	60.4
• DUSA (Ours)	64.1	67.7	68.3	54.8	56.2	64.6	65.6	69.8	69.9	74.5	79.0	70.3	68.5	71.9	70.7	67.7

Fully TTA of ADE20K Pre-trained Segmentors

Method	Noise			Blur				Weather				Digital				Avg.
	Gauss.	Shot	Impul.	Defoc.	Glass	Motion	Zoom	Snow	Frost	Fog	Brit.	Contr.	Elastic	Pixel	JPEG	
Segformer-B5 (LN+BN)	14.2	15.8	15.6	23.1	16.8	22.5	10.3	22.3	21.5	38.6	42.0	23.1	24.5	33.1	35.3	23.9
• BN Adapt	10.8	12.0	11.7	16.6	12.8	16.6	7.9	17.0	16.8	29.6	32.4	18.2	19.2	25.5	26.3	18.2
• Tent	11.2	13.0	12.5	17.0	13.5	16.9	7.7	17.7	17.4	29.7	32.5	18.6	20.0	25.8	26.4	18.7
• CoTTA	14.6	16.1	15.8	22.6	16.5	22.1	9.8	20.9	20.4	38.8	42.3	21.9	24.3	33.6	35.4	23.7
• DUSA (Ours)	23.6 _{±1.3}	24.5 _{±1.0}	23.2 _{±0.3}	24.7 _{±0.5}	23.2 _{±1.2}	24.7 _{±0.6}	12.5 _{±0.6}	27.3 _{±1.2}	26.7 _{±1.2}	39.3 _{±0.2}	42.6 _{±0.3}	27.1 _{±1.2}	30.6 _{±0.6}	35.7 _{±0.7}	35.6 _{±0.7}	28.1



Competitive Performance with Efficient Compute Budget

$b = k + m$ is total budget, k